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Sentiment Analysis of Social Media

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Abstract:

This paper presents an end-to-end sentiment analysis system designed to monitor user-generated content on social media platforms like Twitter (X) and Instagram. Leveraging transformer-based models and a modular Python framework, the system performs real-time sentiment classification, hashtag trend detection, and engagement analytics. The processed data is visualized via dynamic dashboards using Power BI. This multi-platform pipeline helps researchers and marketers understand public opinion shifts and optimize decision-making strategies based on user sentiment.

Keywords — **Sentiment Analysis, Meta, NLP, Machine Learning, Social Media Insights**

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I. INTRODUCTION

Social media is an integral part of modern communication, reflecting real-time public sentiments about brands, events, and societal issues. Platforms like Twitter and Instagram act as live sentiment sensors, capturing user emotions and feedback in their rawest forms. Analysing this data using sentiment analysis can reveal deep insights into user behaviour, perception trends, and the broader socio-political climate.

However, social media data presents challenges: unstructured content, slang, sarcasm, emojis, and platform-specific features make conventional NLP techniques less effective. To address this, we employ a pipeline combining advanced transformers (e.g., RoBERTa), structured trend analysis, and real-time visualizations to decode and present public sentiment in a digestible format.

II. LITERATURE REVIEW

The development of sentiment analysis has paralleled the advancements in natural language processing (nlp). Initially, sentiment classification systems heavily relied on rule-based and lexicon-driven approaches. These techniques employed pre-existing dictionaries like vader, afinn, and sentiwordnet to assign sentiment scores to specific words or phrases. They were praised for their straightforwardness and openness, especially in situations where understanding the reasoning behind decisions was essential. Unfortunately, their limited understanding of subtle emotions, sarcasm, or contextual changes made them inadequate for comprehending complex social media texts, which are typically informal and emotionally charged.

As the limitations of systems that rely solely on word meanings became evident, the field transitioned towards utilizing machine learning techniques.

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Classical models like naive bayes and support vector machines (svms) were widely embraced. These models utilized methods such as term frequency-inverse document frequency (tf-idf) or bag-of-words to transform textual data into numerical feature vectors. While these techniques provided better results than traditional lexicons, they still faced challenges in understanding long-range connections and grasping the contextual meaning of social media posts.

The emergence of deep learning revolutionized sentiment analysis, leading to significant advancements in the field. Rephrase Recurrent neural networks (rnns), and in particular long short-term memory (lstm) models, facilitated improved management of sequential data. These architectural designs took into consideration word order and context, making them more proficient in comprehending intricate sentence structures. Despite this, they needed extensive datasets for training and were computationally demanding, making real-time analysis difficult.

Transformers, especially models like BERT (bidirectional encoder representations from transformers) and RoBERTa, have revolutionized the current state-of-the-art. These models employ attention mechanisms to assign importance to words in a sequence, offering a more comprehensive understanding of context than previous methods. Initially trained on extensive collections of data and subsequently optimized for specific tasks such as sentiment analysis, transformers exhibit exceptional precision and flexibility across diverse domains. In recent times, hybrid approaches have also come into existence, blending rule-based techniques with machine learning or deep learning to strike a balance between interpretability and performance. These models are especially beneficial in situations where resources are limited or when transparency is crucial.

The literature suggests a clear path: from basic rule-based methods to advanced, context-aware deep learning and transformer models. This change has paved the way for advanced applications like sarcasm detection, emotion recognition, multilingual sentiment classification, and real-time sentiment tracking, making sentiment analysis an indispensable tool for comprehending public sentiment in digital spaces.

III. METHODOLOGY

The proposed sentiment analysis system is structured with five main components: data collection, data preprocessing, mood classification, trend analysis, and visualization. Each component is developed independently in Python, guaranteeing scalability and seamless integration with other systems. The system is engineered to handle massive volumes of data in real-time and generate actionable mood insights with exceptional accuracy and user-friendly interfaces.

A. Data Collection

The data collection phase uses custom-built Python modules to extract content from social media platforms, primarily Twitter (X) and Instagram. The TwitterCollector class interfaces with the Twitter API through the Tweepy library to collect tweets filtered by hashtags, keywords, or specific user handles. For Instagram, the system employs Instaloader combined with cookie-based API access to scrape post captions, comments, timestamps, and engagement metrics such as likes and comment counts. Both collectors transform the extracted data into normalized JSON formats, which include metadata like platform source, post ID, and time of posting. This normalization ensures compatibility across analysis modules and simplifies downstream processing.

B. Data Preprocessing

Once the data is collected, it undergoes a preprocessing stage to clean and standardize the textual content. Social media data is inherently noisy, often containing URLs, emojis, mentions, hashtags, and informal language. The preprocessing pipeline, built around the TextPreprocessor class, removes these elements using regular expressions

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and natural language processing techniques. Stopwords are eliminated using the NLTK library, and text is normalized by converting to lowercase and removing punctuation and extra whitespace. Hashtags are extracted separately for trend tracking. Each post is transformed into a clean, tokenized version ready for sentiment classification.

C. Sentiment Classification

The cleaned data is passed through the sentiment analysis engine, which supports two approaches. The primary classifier is based on RoBERTa, a transformer model fine-tuned for social media sentiment analysis using the `cardiffnlp/twitter-roberta-base-sentiment` checkpoint. This model evaluates each post and categorizes its sentiment as Positive, Neutral, or Negative, while also generating

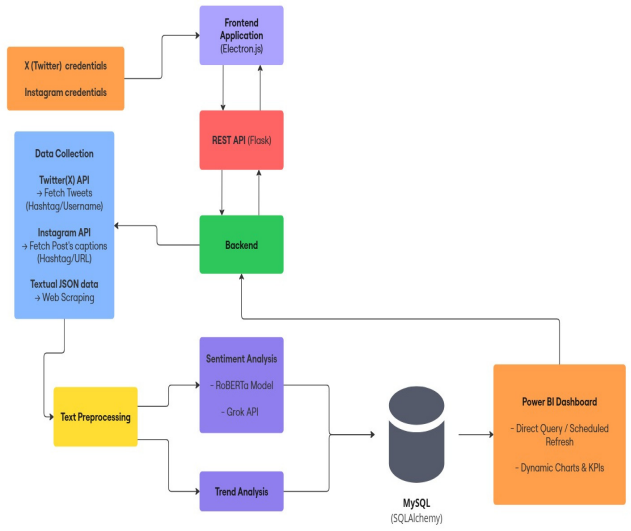


Fig.1

a confidence score. It operates in batches to maximize processing efficiency and includes error handling for cases with empty or malformed input.

An alternative sentiment classification method is implemented using the Grok API, a cloud-based large language model. This approach generates prompts from cleaned text and sends them to a remote inference engine, which returns batch predictions in JSON format. This model is particularly useful in environments with limited GPU resources or where external inference services are preferred.

D. Trend Analysis

After sentiment tagging, the system performs trend analysis to extract insights from the labelled data. The TrendAnalyzer class calculates various metrics, including sentiment distribution across platforms, hashtag popularity, and user engagement levels. It tracks how sentiment changes over time—hourly, daily, or weekly—and correlates these shifts with hashtag frequencies and engagement metrics such as likes, replies, and retweets. Platform-specific metrics are also computed, enabling comparative studies between Twitter and Instagram. The results are compiled into structured reports in JSON format, which can be reused for automated reporting or visualization.

E. Visualization

To make the analysis accessible and dynamic, the final stage involves data visualization using Power BI. The processed data is exported to a MySQL database, which serves as a live source for Power BI dashboards. These dashboards visualize sentiment timelines, top hashtags, platform usage, and engagement KPIs through bar charts, pie graphs, scatter plots, and word clouds. Using Microsoft's Power BI JavaScript SDK, the dashboards are embedded directly into a frontend application. This allows stakeholders to interact with real-time visualizations without needing direct access to Power BI Desktop. The entire setup ensures that backend processing and frontend reporting remain decoupled yet synchronized, supporting real-time decision-making based on user sentiment trends.

II. RESULTS

The implementation of this multi-stage sentiment analysis system provided meaningful insights into public discourse surrounding prominent entities, particularly Meta. By applying RoBERTa-based sentiment classification to a large dataset of social

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media posts collected from Twitter and Instagram, we were able to evaluate both the quality and depth of public sentiment over time. The results revealed a dynamic emotional landscape, where public opinion oscillated in response to events such as product launches, policy announcements, and controversies related to privacy or data security.

Sentiment classification output was categorized into positive, neutral, and negative classes, with corresponding confidence scores. In practice, the RoBERTa model demonstrated strong contextual sensitivity, accurately interpreting short-form, informal, and slang-heavy posts common on social media. During positive sentiment peaks, user reactions aligned with favourable announcements or feature updates from Meta, suggesting strong audience engagement and approval. Conversely, dips in sentiment typically corresponded with negative press or criticism over data practices, often accompanied by spikes in comment volume and hashtag usage.

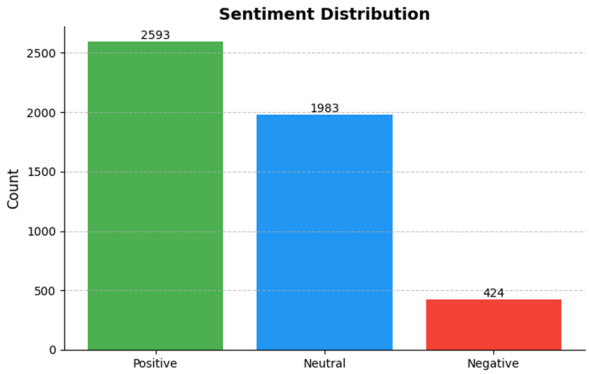


Fig. 2

Further insights were drawn from trend analytics. Hashtag analysis highlighted the most frequently discussed themes, many of which directly mirrored the sentiment scores associated with them. For example, hashtags like #DataPrivacy and #MetaScandal showed a higher proportion of negative sentiment, while hashtags related to innovation or product releases had more positive associations. Engagement metrics—such as average likes, comments, and retweets—were computed across platforms, allowing comparison of user interaction patterns. Interestingly, Instagram posts tended to receive more passive engagement (likes), while Twitter showed higher levels of discussion through replies and quote tweets.

Temporal sentiment analysis provided another layer of understanding. By mapping sentiment scores across hourly, daily, and weekly intervals, the system effectively captured emotional shifts and highlighted the impact of key moments. For example, a sharp increase in tweet volume and negative sentiment was observed following reports of a data breach, while sustained positive sentiment was evident around the release of new features in Meta's VR platform. These visualized results, displayed through embedded Power BI dashboards, facilitated quick interpretation by end-users and stakeholders, enabling real-time reaction to evolving narratives.

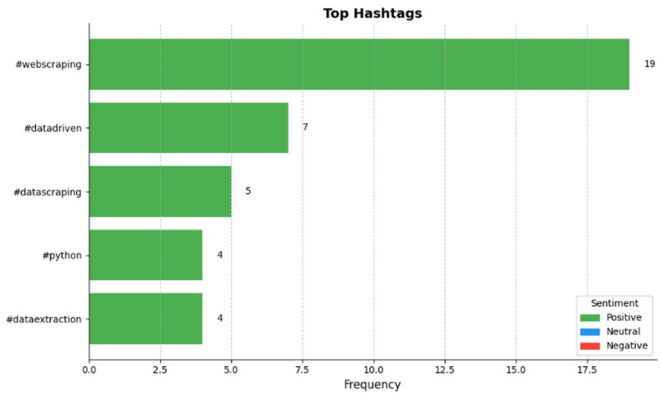


Fig.3

The overall system performance, particularly the RoBERTa classifier, demonstrated notable accuracy and robustness. While some limitations were noted—such as ambiguity in sarcasm or regional language usage—fallback mechanisms and preprocessing steps minimized misclassification. The Grok API-based alternative proved beneficial in scenarios where quick, lightweight analysis was preferred or GPU resources were constrained. These findings affirm the practicality and reliability of integrating transformer-based NLP with real-time visual analytics for scalable sentiment intelligence.

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III. CONCLUSION

This study presents a comprehensive system for performing real-time sentiment analysis on social media content, utilizing a combination of transformer models, structured analytics, and modern visualization tools. By unifying Python-based backend processing with Power BI-based dashboards, the system enables stakeholders to monitor sentiment fluctuations, assess engagement, and detect emerging trends across Twitter and Instagram. The use of a RoBERTa transformer model for sentiment classification significantly enhances the accuracy of emotional interpretation, especially in the informal and dynamic language environment of social media.

The architecture's modularity and extensibility allow it to adapt to additional platforms or evolving analytical needs. Results from this implementation clearly demonstrate the utility of such a system for tracking public opinion, gauging brand reputation, and providing evidence-based decision support. Whether deployed in marketing analytics, public relations, or digital policy assessment, the system empowers users to convert unstructured online data into strategic insights.

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